Induction motor

The motor converts electrical energy into shaft rotation. The frame, stator, rotor, bearings, terminal enclosure, cooling fan, and mounting interface each affect reliable operation. Nameplate voltage, frequency, current, speed, duty, and enclosure ratings are equipment-specific; use the actual nameplate and approved records rather than assumptions.

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Cooling components

The external fan and shroud move air over the motor frame on many designs. Fins increase heat transfer area. Enclosed or specialty motors may use different cooling arrangements. Blocked airflow can contribute to elevated temperature. Confirm design and approved cleaning methods before service; do not remove a guard while running.

Bearing and lubrication

Bearings support the rotor and allow rotation. Bearing arrangement and lubricant requirements vary by design. Incorrect lubricant, excess quantity, contamination, misalignment, and wear can contribute to heat or noise. Confirm bearing type, lubricant, quantity, and interval from the equipment record or approved supplier data.

Belt and pulley drive

Belts transfer rotation between motor and driven equipment. Pulley alignment, belt condition, specified tension, guarding, and load affect service life. Use the drive's documented specification and tools for inspection. A visual check can identify damage but cannot establish correct tension or alignment precisely.

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Electrical terminal enclosure

The terminal enclosure contains connections between supply conductors and motor leads. Connection arrangement and torque are specific to the motor. Only qualified personnel may inspect or service connections using approved isolation and verification procedures. Enclosure appearance alone does not establish electrical safety.

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